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SUMMARY

Data & Cloud engineer, backend developer and applied-AI builder with 3 years of industry experience, shipping cloud infrastructure, data pipelines, backend services and the AI on top, end to end. Built enterprise **ETL pipelines** at Comcast (**50% cost cut, 60% fewer failures**), real-time ML on **17.2M** records at MInt Labs (IIT Madras), and a complete UK cloud traceability platform at Kahawa Roots. Completing a Joint MSc in Data Science & AI (University of Birmingham & IIT Madras). Open to data engineering, cloud/platform/DevOps, backend and applied-AI roles across the UK, EU and Dubai (remote-friendly; UK Skilled Worker sponsorship required for long-term UK roles).

EXPERIENCE

Kahawa Roots (UK) Ltd [reference](#)

May – Jul 2026

Traceability Platform Developer Intern

United Kingdom

- Designed and delivered an enterprise **coffee traceability platform** (Kenyan farms to UK buyers and consumers) end to end – system architecture, workflows and data flow.
- Built **REST APIs in Python/FastAPI** on a **PostgreSQL** schema – data modelling and migrations for lot tracking, record correction and approval workflows.
- Developed the **Next.js** frontend, including customer-facing **QR traceability journeys**.
- Provisioned the full **AWS** stack (**ECS, RDS, S3, Cognito**) as **Terraform** infrastructure-as-code with **GitHub Actions** **CI/CD** and **OIDC**; implemented **Cognito authentication, RBAC, audit logging and secrets management**.
- Ran regression/deployment testing, production monitoring and incident debugging; technical lead with stakeholders and mentored a fellow intern – recognised in a reference letter from the Head of Sustainability & Traceability.

MInt Labs, IIT Madras [reference](#)

Nov 2025 – Jan 2026

Winter Project Intern [reference](#)

Chennai, India

- Built a **real-time bus ETA system** for the IITM campus shuttle from streaming GPS: **17.2M records, 53 spatial-temporal features**, and a **GNN + XGBoost + LightGBM** ensemble served via **FastAPI + Docker**.
- Delivered **1.88 min MAE** with sub-millisecond inference and **100% uptime**; built validation, monitoring and fallback strategies for robust performance under noisy GPS data.

Comcast [reference](#)

Jul 2023 – Jun 2025

Development Engineer 1, Data Engineering

Chennai, India

- Built and optimised enterprise **ETL pipelines** with **Kafka, Redshift and PySpark**, ingesting sources such as **Workday and LDAP** into the enterprise data warehouse.
- Designed automated **data reconciliation frameworks** validating source-to-warehouse integrity, improving reporting accuracy by **40%**.
- Migrated legacy reporting jobs to modern pipelines, reducing failures and execution time by **60%**.
- Reduced infrastructure costs by **50%** through warehouse consolidation and optimised snapshot/storage strategies.
- Built **EnvSyncer** for cross-environment data sync, plus an **anomaly-detection PoC (AWS Lookout for Metrics)** wired into **CloudWatch** dashboards and **Slack** alerting.
- Led **knowledge transfer** and onboarding docs; supported the **Service Reliability Engineering** team on deployment strategy via testing and performance tuning.

Comcast

Jan 2023 – Jun 2023

Intern, Data Engineering

Chennai, India

- Developed a **Slack monitoring system** for ETL runs and job status, providing real-time alerts and daily cost breakdowns.
- Contributed to the enterprise **data warehouse** application – a single source of truth consolidating multiple data sources.

EDUCATION

University of Birmingham & Indian Institute of Technology Madras

Jul 2025 – 2027 (expected)

Joint MSc in Data Science and Artificial Intelligence

Birmingham, UK & Chennai, India

- Coursework: Machine Learning, Data Analytics, NLP, Data Visualisation, DSA & Mathematics for Data Science.

SASTRA Deemed University [reference](#)

Jul 2019 – May 2023

Bachelor of Technology in Computer Science & Engineering

Thanjavur, India

- First Class with Distinction**; core-CS foundation in data structures, algorithms, DBMS and operating systems.

PROJECTS

DeltaEd – Growth-Based Student Leaderboard & AI Coach

Google Hackathon

- Built a **RAG-powered AI coach** that ranks students by how much they personally improve rather than by absolute standing, deployed on **Google Cloud Run**.
- Owned the leaderboard and the retrieval-augmented coaching loop that walks each student through their gaps; 🏆 won **Best Pitch** and **Best Use of Google Tools**.

Sentiment Analysis & Topic Modeling for Tweets

2024

- Applied **pysentimiento** and **cardiffnlp** NLP models to tweet sentiment and topics (**94.53% mean probability**); built engagement metrics and heat-map visualisations; presented at the **G20 Library Summit**.

Diabetes Prediction Web App

2022

- Built a **Django** app comparing **six ML models** to predict diabetes risk (**Random Forest, 0.87 accuracy**), generating downloadable **PDF reports** and **SMS notifications**.

VAC – Video As Code

Personal project

- Building a purely **algorithmic compiler/decompiler between video and human-readable source code** (pixels to code, measure-don't-classify).

Other: [ACEest Fitness \(Flask + Docker + CI/CD\)](#), [IITM MLP Project \(Kaggle, 87% accuracy\)](#), [Metric-Learning Data Challenge \(Kaggle\)](#), [Helper Libraries \(Python/AWS\)](#), [IITM Business Data Management](#)

TECHNICAL SKILLS

- **Languages:** Python, SQL
- **Backend:** FastAPI, Django, Flask, REST APIs, SQLAlchemy, boto3, Pandas
- **Frontend:** Next.js, React
- **Cloud:** AWS (ECS, RDS, S3, Cognito, Lambda, Glue, Athena, Redshift, IAM, CloudWatch, Secrets Manager, Lookout for Metrics), GCP (Cloud Run)
- **DevOps & IaC:** Terraform, Docker, GitHub Actions, Jenkins, CI/CD, OIDC, Git, Linux, Bash
- **Data Engineering:** ETL/ELT, PySpark, Kafka, Data Warehousing, Data Modelling, PostgreSQL, Redshift, PyArrow
- **AI/ML:** Machine Learning, XGBoost, LightGBM, Graph Neural Networks, RAG, NLP
- **Security & Testing:** IAM, RBAC, authentication, audit logging, secrets management, Pytest, regression & deployment testing

AWARDS & ACHIEVEMENTS

- 🏆 **Best Pitch** and **Best Use of Google Tools** – Google Hackathon (DeltaEd).
- **Champ Award**, Comcast (Q4 2023, Q2 2024) – for consecutive outstanding performance.
- **Good Presentation Award**, **G20 Library Summit**  (Feb 2024).
- **First Class with Distinction**, B.Tech (SASTRA); MSc passed with distinction.

WORKSHOPS, TALKS & CONFERENCES

- **G20 Library Summit**  (Feb 2024) – presented sentiment analysis and topic modelling on X/Twitter data.
- Presented data engineering solutions at **Comcast** (2024) – performance optimisations and key insights.
- Organised a **SQL Workshop Fundraiser**  for **Bhumi** (Sep 2024) – raised funds to support shelter children.
- Attended the **Responsible & Safe AI Workshop** at **IIT Madras**  (Nov 2023) – legal bias, AI attacks and unlearning (CeRAI).
- **AWS User Group Chennai**  (Sep 2024) – GenAI and serverless, with speaker Eric Johnson.
- **IIT E-Summit 2024**  (Mar 2024) – talks by venture capitalists on startups and investing.
- **AWS Awesome Day**  (2023) – machine learning, serverless and big-data analytics.
- **Web Development Workshop**, SASTRA (Aug 2022) – three-day hands-on Django/Python program.

VOLUNTEERING & EXTRACURRICULARS

Bhumi

Jul 2024 – Present

Centre Coordinator (Sep 2024 – Present) · LEM Volunteer (Jul – Sep 2024)

Chennai, India

- Organised workshops, events and additional classes for foster children; ran sports events to raise funds.
- Taught mathematics to shelter-home children via the Ignite **Little Einstein Mathematics** program, using interactive hands-on methods.

SASTRA – National Service Scheme (NSS)

Aug 2019 – Jun 2024

NSS Leader (2022–24), Volunteer (2019–22)

Thanjavur, India

- Increased annual blood-donation camps and led crowd control and festival management.
- Strengthened community ties through hospital support, temple ethics drives and old-age-home cleanups.